1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

In muscle-invasive urothelial carcinoma treated with a pembrolizumab-containing regimen following cisplatin-based chemotherapy, cisplatin-induced DNA damage increases neoantigen burden and recruits tumor-infiltrating lymphocytes (TILs). However, loss-of-function TP53 alterations impair the effective translation of this augmented immunogenicity into productive anti-tumor T cell responses by promoting defects in antigen presentation and/or establishing a pro-immunosuppressive cytokine milieu. Consequently, high TIL density will predict improved pembrolizumab response primarily in TP53-wildtype tumors, whereas TP53-mutant tumors will show attenuated or absent benefit from pembrolizumab despite comparable or higher TIL infiltration. This effect is modulated by secondary factors such as baseline tumor mutational burden (TMB), ERCC2/other DDR alterations, and the quality (clonality) of neoantigens. The testable predictions include: (1) a significant interaction between TP53 status and pretreatment TIL density predicting pembrolizumab response; (2) TP53-wildtype, high-TIL tumors will have the highest response rates; (3) TP53-mutant tumors will exhibit reduced response regardless of TIL density due to impaired antigen presentation and/or immunosuppressive signaling; (4) biomarkers of antigen presentation (MHC I/B2M/TAP1/2) and interferon signaling will mediate the TP53–TIL–response axis and correlate with outcome.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	TP53
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

TP53: TP53 status; why do some tumors with apparent immune infiltration nevertheless fail to respond to PD-1 blockade? 2) Tier 1 path used: the single path linking cisplatin → induced genetic mutations → TP53 alterations → urothelial carcinoma, plus the clinical trial context of pembrolizumab; I built on TP53 as the primary molecular anchor because it is central to DDR, genomic instability, and tumor biology in MIBC.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	Cisplatin	Supporting modifier from knowledge graph

Secondary	Tumor mutational burden / neoantigen load	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- Cisplatin: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- Tumor mutational burden / neoantigen load: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.